Title to be chosen

2026-11-14

Introduction

Plan

- **The gap.**
  - Real-time case counts are incomplete because of reporting delays.
  - Generic forecasting models do not use the reporting structure.
  - Key references: (Höhle and Heiden 2014; Johnson et al. 2025; Reich et al. 2022).
- **Context.**
  - Nowcasting methods model reporting delays directly.
  - Time series foundation models forecast well with no training on the target data.
  - Training on domain-specific synthetic data has precedent: KernelSynth for Chronos (Ansari et al. 2024), ForecastPFN (Dooley et al. 2023), prior-fitted networks (Müller et al. 2022; Hollmann et al. 2025), and Mantis for outbreaks (Dudley et al. 2026).
  - Key references: (McGough et al. 2020; Günther et al. 2021; Wolffram et al. 2023; Ansari et al. 2024; Das, Kong, et al. 2024; Woo et al. 2024; Rasul et al. 2023; Garza, Challu, and Mergenthaler-Canseco 2023; Aksu et al. 2024; Shchur et al. 2025; Kalahasti et al. 2026; Dudley et al. 2025).
- **What we did.**
  - We fine-tuned a foundation model on synthetic outbreak data, following that precedent but fine-tuning a general model rather than training from scratch.
  - We evaluated it retrospectively and prospectively.
  - Key references: (Ansari et al. 2025; Johnson et al. 2025).

Methods

Data

> **Plan**
>
> - The German COVID-19 hospitalisation nowcast hub.
> - The Ebola forecasting hub target and its releases.
> - Key references: (Wolffram et al. 2023; KIT Metrics Lab et al. 2025; InsightNet-US 2026; MacNeil et al. 2010).

Baselines

> **Plan**
>
> - `randomwalk`, a random walk with no delay adjustment, and `randomwalknowcast`, a random walk nowcast.
> - Both in full in the SI.
> - Key references: (Johnson et al. 2025).

Foundation model setup and fine-tuning

> **Plan**
>
> - Chronos-2, given the reporting triangle as series: one series per place and reporting delay, each missing until reported. Introduced alongside the reporting triangle, with Figure 1B.
> - Fine-tuned on synthetic data on a laptop, with checkpoints after 1,000, 2,000, 4,000, 6,000 and 8,000 simulated outbreaks. The largest is the main model.
> - Compared with Chronos-2 given only complete data points, to show what the ragged structure and fine-tuning add.
> - Figure 1, a multi-panel schematic of our pipeline: (A) the assumptions in the epinowcast simulations, (B) the reporting triangle laid out as ragged series for Chronos-2, (C) training blocks and LoRA fine-tuning, and why it is fast, and (D) forecasting with the fine-tuned model.
> - Key references: (Ansari et al. 2025, 2024; Hu et al. 2021; Das, Faw, et al. 2024).

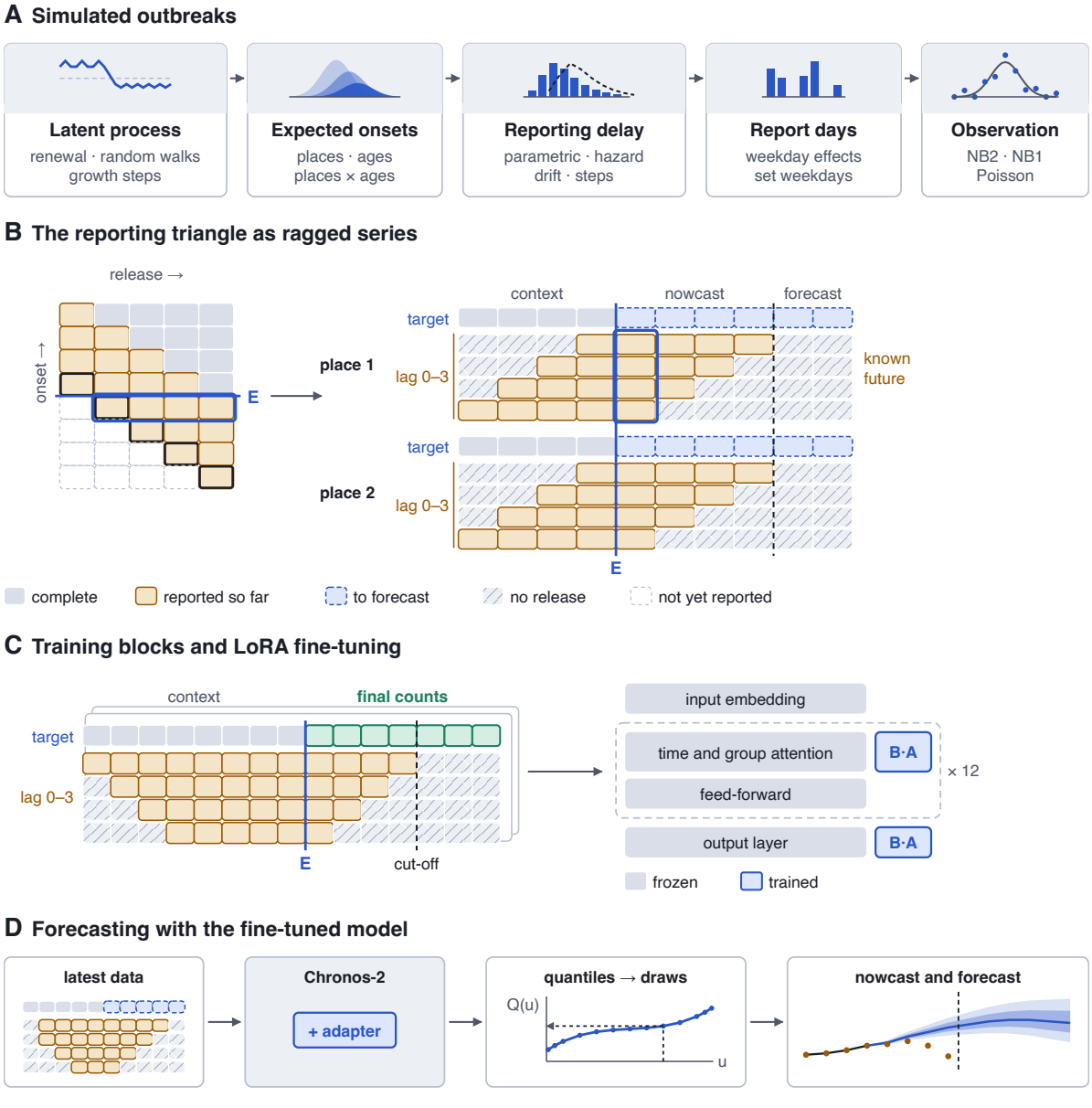


Figure 1: The fine-tuning pipeline. (A) The assumptions in the epinowcast simulations: a latent process gives expected onsets by stratum, which a reporting delay that may drift or step, report-day effects and count noise turn into a reporting triangle. (B) The reporting triangle laid out as ragged series for Chronos-2: each place has a target complete up to E and one series per lag, missing until reported (daily lag series fill only on release days), and counts reported after E enter as known-future inputs. (C) Training blocks and LoRA fine-tuning, and why it is fast: blocks cut from simulated release histories at random cut-offs are scored against the final counts, and only rank-8 adapters, about 1% of the weights, are trained. (D) Forecasting with the fine-tuned model: one forward pass gives quantiles for each step, and each draw reads them at a uniform level u and is at least the count reported so far.

Synthetic data generation

Plan

- Simulated with epinowcast over a grid of outbreak, delay and noise settings.
- A table in the main text gives the grid of synthetic scenarios.
- The full process and an overview of the epinowcast model used are in the SI.
- Key references: (Abbott et al. 2026; Lison et al. 2024; Park et al. 2024; Ansari et al. 2024; Dooley et al. 2023; Müller et al. 2022; Hollmann et al. 2025).

Table 1: Placeholder: grid of synthetic scenarios.

Setting	Values
Outbreak	
Delay	
Noise	

Evaluation on held-out synthetic data

Plan

- A held-out set of simulations was not used during development.
- Sample forecasts scored with CRPS and log-scale CRPS, bias, and coverage, relative to `randomwalknowcast`.
- Scored against a rolling evaluation set of the data as reported D days later, with D each outbreak's 98% delay.
- Key references: (Bracher et al. 2021; Bosse et al. 2023; Gneiting and Raftery 2007; Gneiting, Balabdaoui, and Raftery 2007; Sherratt et al. 2023).

Case study 1: German nowcast hub

Plan

- Compared with the hub's models, relative to the hub's own baseline, and evaluated as the hub did.
- Quantile forecasts scored with WIS and log-scale WIS, bias, and coverage, against a rolling evaluation set of the data as reported D days later (Wolffram et al. 2023).
- Daily and weekly versions may not both make sense here.

- The nowcast dates are split in half. We developed our setup on the first half only, and the second half is held out for the results.
- Key references: (Wolffram et al. 2023; Johnson et al. 2025; Amaral et al. 2025).

Case study 2: prospective submission to the Ebola forecasting hub

i Plan

- Compared with the other submissions, relative to the hub's own baseline.
- Scored as for the held-out synthetic data, with D the empirical 98% point of the hub data's reporting delay.
- Our models' forecasts are saved prospectively, alongside the main submissions as they were made.
- Key references: (InsightNet-US 2026; Viboud et al. 2018; Funk et al. 2019).

Implementation

i Plan

- The tools we used, each cited: Python, PyTorch on the laptop GPU, chronos-forecasting, PEFT for LoRA, R and epinowcast for simulation, and scoringrules for scoring, checked against scoringutils.
- The evaluation runs as a Snakemake workflow, so every result can be reproduced from the raw data.

Results

i Plan

Each section has a figure panel. Each panel shows:

- nowcasts over time (ribbons);
- scores relative to the baseline (`randomwalknowcast` for synthetic data, the hub's baseline for each case study), with natural- and log-scale WIS (or CRPS, MAE where they make sense);
- bias;
- 50% and 90% coverage;
- score variation by location and horizon, supported by SI plots.

- for Germany, a dotted line marking the split between the development and held-out halves, with scores split by half.

The plots in primarycensored-paper are a model for these. Each section has two paragraphs, one on visual performance and one summarising scores.
Key references: the Methods references, with (Bracher et al. 2021; Bosse et al. 2023) for the natural- and log-scale panels.

Held-out synthetic data

Plan

- Visual performance of the candidate models against `randomwalk` and `randomwalknowcast`.
- A summary of score performance.
- How scores change with the amount of fine-tuning data (1,000 to 8,000 outbreaks), as an SI figure.

Figure to come.

Figure 2: Placeholder: held-out synthetic data results panel.

German nowcast hub

Plan

- Visual performance against the other models.
- A summary of score performance.
- Key references: (Wolffram et al. 2023; Amaral et al. 2025).

Figure to come.

Figure 3: Placeholder: German nowcast hub results panel.

Ebola hub submissions

Plan

- Visual performance against the other submissions.
- A summary of score performance.

Figure to come.

Figure 4: Placeholder: Ebola hub submissions results panel.

Discussion

Plan

- **Summary** of what we did.
- **Strengths and weaknesses.**
  - Weaknesses: only one foundation model, only one source of synthetic data, and limited case studies.
  - Weakness: we did not compare against epinowcast, or, for the synthetic study, against the data-generating model itself.
  - Strengths: both retrospective and real-time case studies, and synthetic data that are easy to tune to a setting.
  - Strength: fine-tuning ran locally on consumer hardware (a laptop), so it is accessible.
  - Weakness: fine-tuning used the laptop's GPU, which made it several times faster, and not everyone has one. On a laptop CPU alone it would still run in hours.
  - Key references: (Kalahasti et al. 2026; Meyer et al. 2025).
- **Domain knowledge through synthetic data.**
  - Domain knowledge enters through the synthetic data source.
  - Fingers crossed it improves results; to fill in from the results.
  - Key references: (Dudley et al. 2026, 2025; Dooley et al. 2023; Hollmann et al. 2025; Ansari et al. 2025).
- **Against the literature.**
  - Other uses of foundation models in infectious disease modelling. Key references: (Kalahasti et al. 2026; Cai, Huang, and Xu 2026; Wang et al. 2025; Jafari et al. 2026; Du et al. 2025; Li et al. 2025; Aygün et al. 2025; Martinson et al. 2026).

  - Fine-tuning in other domains. Key references: (Laglil et al. 2026; Das, Faw, et al. 2024; Hu et al. 2021).
  - Other nowcasting approaches. Key references: (Bastos et al. 2019; Seaman et al. 2022; Bergström et al. 2022; Sumalinab et al. 2024; Schneble et al. 2021; Kassteele, Eilers, and Wallinga 2019; Overton et al. 2023; Mellor et al. 2025; Tang et al. 2025; Stoner, Economou, and Halliday 2020; Kamarthi, Rodríguez, and Prakash 2021; Koemen et al. 2025; Wilsens, Antonio, and Claeskens 2025).

- **A possible emulator.**
  - Fine-tuning on epinowcast simulations may give a fast emulator of that package.
  - We did not compare against epinowcast, so we cannot say how close it is.
  - This could be useful for other slow, long-running packages.

- **Further work.**
  - Other foundation models.
  - More case studies.
  - A comparison with custom neural networks.
  - Testing the fine-tuned model as an emulator of epinowcast.
  - Fine-tuning with other epidemiological features as covariates, such as distances between locations or mobility.
  - Key references: (Das, Kong, et al. 2024; Woo et al. 2024; Rasul et al. 2023; Kamarthi, Rodríguez, and Prakash 2021; Koemen et al. 2025; KIT Metrics Lab, Bracher, and Schienle 2025).

- **Conclusions.**
  - Reuse state-of-the-art generic forecasting models with domain-specific insight.
  - Fine-tuning on a package's simulations could give a fast emulator of that package, which may help other long-running packages.
  - It shows promise, and more work is needed.

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Data and software availability

AI disclosure

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